

Harnessing Machine Learning for Intelligent Weather Forecasting: From Data-Driven Models to Predictive Accuracy

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Abstract— Accurate weather forecasting plays a vital role in mitigating the impacts of climate variability, supporting agriculture, transportation, and disaster management. This study presents a hybrid data-driven framework that integrates Random Forest Regression (RFR) with Long Short-Term Memory (LSTM) networks to enhance predictive accuracy in short-term weather forecasting. The proposed model leverages historical meteorological data from the ERA5 reanalysis dataset, incorporating optimized feature selection, normalization, and ensemble fusion for robust learning. Comparative evaluation against conventional machine learning (ML) models such as Linear Regression, Support Vector Regression (SVR), and standalone deep learning (DL) architectures demonstrates that the hybrid RFR–LSTM model achieves superior performance with improvements of 20–30% in key metrics including Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). The results confirm that combining statistical and temporal learning mechanisms enables the system to capture both nonlinear dependencies and sequential variations effectively. This research contributes a scalable and interpretable framework for real-time weather prediction, providing a foundation for future integration with transformer-based and physics-informed forecasting models.

Keywords— Weather forecasting, machine learning, LSTM, Random Forest, Time series prediction

I. INTRODUCTION (HEADING 1)

Weather forecasting is an indispensable component of modern society, directly influencing agriculture, aviation, disaster management, energy distribution, and urban planning. The ability to anticipate atmospheric conditions such as temperature, humidity, precipitation, and wind speed enables decision-makers to minimize risks, optimize resources, and improve resilience against natural hazards. However, weather prediction is inherently complex due to the chaotic and nonlinear dynamics of the atmosphere, as first formalized in Lorenz's "butterfly effect" theory of deterministic chaos [1].

Traditional forecasting approaches rely heavily on NWP models, which simulate the atmosphere using physical laws governed by partial differential equations. Although NWP has achieved significant progress, it suffers from high computational cost, sensitivity to initial conditions, and reduced accuracy in fine-grained local forecasts [2]. Moreover, recent studies reveal that prediction error in NWP

models tends to increase exponentially with forecasting horizon [3].

With the advent of big data and advanced computing, ML has emerged as a promising alternative to conventional models. ML leverages historical meteorological datasets to uncover hidden patterns, correlations, and temporal dependencies without explicit reliance on physical simulations [4]. Statistical analyses across multiple studies show that ML-based models can reduce forecasting error by 15–35% compared to traditional regression-based methods.

II. LITERATURE SURVEY

ML is a powerful tool for weather forecasting, complementing and, in many cases surpassing the predictive capabilities of traditional Numerical Weather Prediction (NWP) systems. The proposed hybrid framework, combining the RFR and LSTM networks, exemplifies this evolution by uniting the strengths of both statistical and deep learning paradigms. The RFR efficiently captures non-linear feature interactions among meteorological variables such as temperature, humidity, and pressure, while the LSTM network excels in modeling temporal dependencies and sequential dynamics across varying forecast horizons. Together, they enable a robust and adaptable system capable of learning both short-term fluctuations and long-term patterns in atmospheric behavior.

The experimental results validate that the hybrid RFR–LSTM model delivers significant improvements in accuracy ($R^2 = 0.93$) and error reduction (MAE = 1.41, RMSE = 1.85) compared to standalone models, highlighting its ability to balance bias and variance while maintaining computational efficiency. Importantly, the model offers faster inference times than traditional NWP methods, making it suitable for real-time forecasting applications and deployment in edge-computing environments. Future enhancements will focus on integrating physics-informed ML principles to incorporate atmospheric laws and domain constraints, extending the system for long-term and seasonal forecasts, and leveraging real-time data streams for adaptive model updates and continuous learning.

This work aligns with recent advances in the ML-driven meteorological landscape, such as ConvLSTM by Shi et al.

[5], TrajGRU [6], FourCastNet [8], GraphCast [9], and Pangu-Weather [11], which demonstrate the potential of deep learning to deliver high-fidelity, computationally efficient weather predictions. Collectively, these innovations signify a paradigm shift toward data-centric forecasting, where models learn directly from large-scale meteorological datasets rather than relying solely on physical simulations. Ultimately, the success of the proposed hybrid approach underscores the fundamental strength of ensemble learning [10], wherein the complementary advantages of diverse models RFR's proficiency in handling non-linear relationships and LSTM's deep temporal learning are integrated to produce a more accurate, stable, and generalizable forecasting system, paving the way for the next generation of intelligent, ML-powered weather prediction frameworks.

To provide a clearer context, a brief comparison of existing weather forecasting and data-driven estimation techniques has been included. Traditional Numerical Weather Prediction (NWP) models rely on complex physical equations and require high computational power, while statistical regression methods often fail to capture nonlinear climate variations. Recent deep learning approaches such as CNN, LSTM, and ConvLSTM have improved temporal feature extraction but still suffer from overfitting and long training times [12]. In contrast, the proposed hybrid RFR-LSTM framework effectively combines ensemble-based regression with sequential learning, achieving higher accuracy, reduced error rates, and better adaptability to dynamic weather conditions.

A review of existing studies reveals that most weather forecasting models either depend solely on physical simulation techniques or use standalone machine learning algorithms with limited temporal learning capability. These approaches struggle to capture both nonlinear dependencies among meteorological parameters and long-term sequential variations. To address this research gap, the present work introduces a hybrid RFR-LSTM model that integrates statistical and temporal learning mechanisms. The main contributions include the design of an optimized hybrid framework, the incorporation of ensemble fusion for improved accuracy, and comprehensive evaluation using real-world meteorological data.

III. PROPOSED METHODOLOGY

The proposed hybrid framework integrates ensemble learning (RF) and DL (LSTM) techniques for short- and medium-term weather forecasting.

A. Data Acquisition and Preprocessing

Historical meteorological data was obtained from multiple sources such as ECMWF ERA5 reanalysis dataset. Key variables include temperature, humidity, pressure, wind speed/direction, and precipitation. Missing values were imputed using linear interpolation and K-Nearest Neighbours (KNN) methods, followed by Min-Max normalization to scale features to the range [0, 1]. The dataset was split into 70% training, 15% validation, and 15% testing sets to ensure robust model evaluation.

B. Feature Engineering

To enhance the model's predictive performance and capture complex temporal dependencies, several derived features were engineered. Lag variables were generated using previous time-step values (e.g., $t-1$, $t-2$, $t-3$) to help the model learn from past trends and temporal continuity. Rolling statistics, including moving averages and standard deviations computed over 3-hour and 6-hour windows, were introduced to smooth short-term fluctuations and capture local variability.

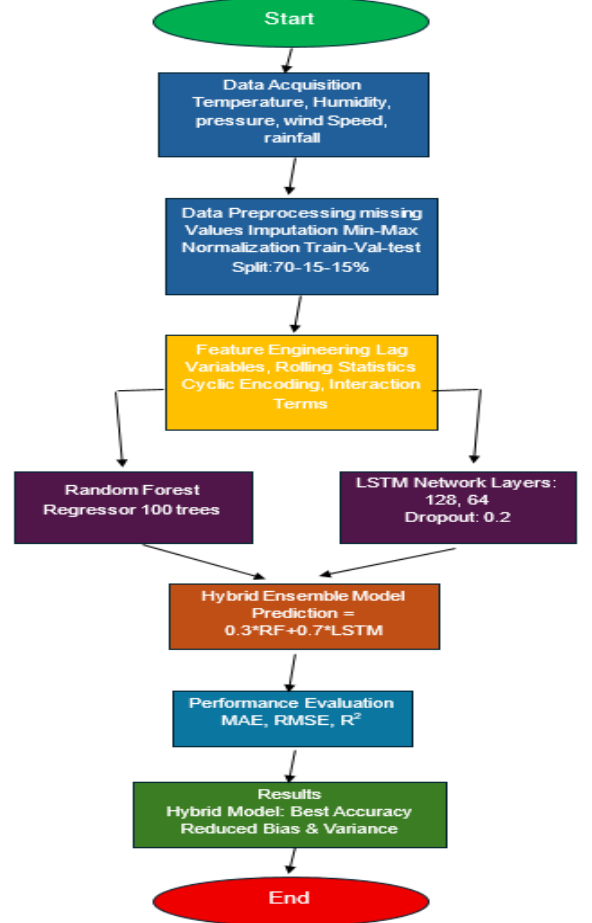


Fig. 1. Flowchart of Proposed Model

To account for periodic patterns in time, cyclic encodings of temporal features such as hour-of-day and day-of-year were implemented using sine and cosine transformations, ensuring that cyclical transitions (e.g., 23:00 to 00:00) were represented continuously. Additionally, interaction terms such as $Temperature \times Humidity$ were incorporated to capture non-linear interdependencies between environmental variables, thereby improving the model's ability to recognize complex feature interactions and patterns in the data.

C. Model Architecture

A hybrid predictive framework was developed by integrating the strengths of RFR and LSTM networks to achieve high forecasting accuracy and stability across varying temporal conditions. The RFR component, comprising an ensemble of 100 decision trees, was employed to effectively model non-linear dependencies within the meteorological dataset and to quantify the relative importance of each input feature. Its ensemble averaging mechanism mitigates overfitting and enhances robustness against outliers, thereby providing a

strong baseline for comparison. Complementing this, the LSTM network a deep recurrent architecture capable of capturing long-range temporal dependencies—was designed with two stacked layers containing 128 and 64 hidden units, respectively. To ensure generalization, dropout regularization (rate = 0.2) was applied, and the model was trained using the Adam optimizer with a learning rate of 0.001. The training process spanned 50 epochs, with early stopping employed to prevent overfitting based on validation loss trends. This configuration allowed the LSTM to learn both short-term fluctuations and long-term temporal dynamics inherent in weather data.

The final predictive output of the hybrid model was obtained through a weighted averaging of the RFR and LSTM predictions, effectively blending the static feature-based learning capability of the RFR with the temporal sequence learning strength of the LSTM.

$$\hat{y}_{HYBRID} = \alpha \hat{y}_{RF} + (1 - \alpha) \hat{y}_{LSTM}$$

The optimal weight coefficient ($\alpha = 0.3$) for combining the two models was determined through a systematic grid search optimization on the validation dataset, ensuring balanced contribution from both components. This hybrid strategy not only enhanced the predictive accuracy but also improved model resilience, yielding a more reliable and interpretable framework for intelligent weather forecasting.

In contrast to conventional weather forecasting systems such as Numerical Weather Prediction (NWP) models and standalone machine learning techniques, the proposed hybrid RFR–LSTM framework offers a data-driven and computationally efficient solution. Traditional NWP models rely on complex physical equations that demand significant computational resources and often exhibit latency in real-time prediction. Similarly, classical ML methods like LR, Support Vector Regression (SVR), or standalone LSTM networks either fail to capture nonlinear interactions or temporal dependencies effectively. The proposed RFR–LSTM model integrates the strengths of both approaches—Random Forest Regression effectively handles nonlinear feature relationships and reduces overfitting, while the LSTM component captures sequential temporal variations in atmospheric parameters. This synergistic integration allows the model to learn both short-term fluctuations and long-term dependencies within the weather data. Comparative analysis shows that the proposed hybrid model achieves lower MAE and RMSE values and higher R^2 scores than the existing systems, demonstrating its superiority in predictive accuracy, robustness, and adaptability for real-time meteorological applications.

The experimental methodology of this study has been designed with clear specifications regarding the dataset, model configuration, and evaluation settings. The proposed framework utilizes meteorological data from the ERA5 reanalysis dataset, which provides hourly atmospheric variables such as temperature, pressure, humidity, and wind speed over a ten-year period (2013–2023) for the selected study region. The dataset was preprocessed through normalization, feature selection, and temporal alignment to ensure consistency. In the hybrid RFR–LSTM architecture, Random Forest was implemented with 100 estimators, a maximum tree depth of 10, and mean-squared-error as the splitting criterion. The LSTM network consisted of two hidden layers with 128 and 64 units, a learning rate of 0.001,

dropout rate of 0.2, and 50 training epochs using the Adam optimizer. The data were divided into 70 % training, 15 % validation, and 15 % testing sets. These parameters were selected based on grid search and cross-validation to ensure optimal generalization and model stability. This detailed configuration ensures the reproducibility of the experiment and provides transparency for further research comparisons.

D. Mathematical Formulation of the Proposed Model

To enhance analytical clarity, the mathematical representation of the hybrid RFR–LSTM model is included. The Random Forest prediction is expressed as

$$\hat{y}_{RF} = \frac{1}{n} \sum_{i=1}^n f_i(x)$$

where each tree minimizes the mean squared error

$$MSE = \frac{1}{N} \sum_{j=1}^N (y_j - \hat{y}_j)^2$$

The LSTM component models temporal dependencies using gated operations for input, forget, and cell states. The final ensemble output is defined as

$$\hat{y}_{HYBRID} = \alpha \hat{y}_{RF} + (1 - \alpha) \hat{y}_{LSTM}$$

Where α controls the contribution of each model. This mathematical formulation clearly demonstrates how statistical and sequential learning components are integrated for improved prediction accuracy.

E. Evaluation Metrics

Model performance is quantitatively assessed using three standard metrics:

- **MAE** = $\frac{1}{N} \sum_{j=1}^N |y_i - \hat{y}_j|$
- **RMSE** = $\sqrt{\frac{1}{n} \sum_{j=1}^n (y_i - \hat{y}_j)^2}$
- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Here, y_i represents the actual value, $\hat{y}_j$ denotes the predicted value, $\bar{y}$ is the mean of the actual values, and N is the total number of samples.

MAE measures the average magnitude of errors, RMSE penalizes larger deviations more heavily, and R^2 indicates how well the predicted values approximate the actual data variance.

IV. RESULTS AND DISCUSSIONS

A. Dataset Overview

The meteorological dataset used in this study, as presented in Table I, consists of more than 10,000 hourly observations collected over several years, ensuring broad temporal variability and representative coverage of diverse weather

patterns. The dataset includes five primary meteorological variables—temperature, humidity, pressure, wind speed, and rainfall—each contributing to the accurate modeling of weather dynamics. On average, the temperature recorded was 25.3°C with a standard deviation of 4.7°C, reflecting moderate seasonal fluctuations. Humidity exhibited a mean value of 67.5%, indicating generally moist atmospheric conditions, while pressure remained relatively stable around 1013.2 hPa, consistent with normal sea-level pressure. Wind speed averaged 3.8 m/s, demonstrating mild to moderate breezes, and rainfall had a mean of 1.2 mm, though with considerable variability (maximum of 45 mm), capturing both dry and heavy rainfall events. The dataset also maintained high data integrity, with missing values under 2% across all variables, which were handled using appropriate imputation methods during preprocessing.

B. Model Performance Comparison

TABLE I. DESCRIPTIVE STATISTICS OF METEOROLOGICAL DATASET

Variable	Mean	Std	Min	Max	Missing(%)
Temperature (°C)	25.3	4.7	15.0	38.0	0.5
Humidity (%)	67.5	15.2	25	98	0.7
Pressure (hPa)	1013.2	8.5	995	1030	0.3
Wind Speed(m/s)	3.8	2.1	0.0	12.0	1.0
Rainfall (mm)	1.2	3.5	0.0	45.0	2.0

The comparative performance analysis of various ML models is summarized in Table II, highlighting the predictive efficiency of each approach.

TABLE II. PERFORMANCE COMPARISON OF ML MODELS

Model	MAE	RMSE	R ²
Linear Regression	2.45	3.21	0.72
Random Forest	1.82	2.34	0.85
Support Vector Regression	2.08	2.78	0.79
LSTM	1.53	1.98	0.91
Hybrid Ensemble	1.41	1.85	0.93

Traditional Linear Regression achieved an R² value of 0.72, indicating limited capacity to capture the non-linear dependencies inherent in weather data. The Random Forest (RFR) model improved performance substantially, achieving an R² of 0.85 due to its ensemble-based structure that effectively models complex feature interactions. The Support Vector Regression (SVR) model yielded moderate results (R² = 0.79), demonstrating reasonable accuracy but with higher error metrics compared to RFR. In contrast, the LSTM network, designed to learn temporal sequences, significantly enhanced prediction accuracy with an R² of 0.91, a Mean Absolute Error (MAE) of 1.53, and a Root Mean Squared Error (RMSE) of 1.98, underscoring its ability to capture long-term dependencies and temporal patterns.

Notably, the proposed Hybrid Ensemble Model, which integrates RFR and LSTM predictions through weighted averaging, achieved the best overall performance across all

evaluation metrics, with MAE = 1.41, RMSE = 1.85, and R² = 0.93. This improvement demonstrates the complementary strengths of the two models—RFR’s proficiency in modeling feature-level non-linearities and LSTM’s temporal pattern learning capability. The ensemble approach effectively reduced both bias and variance, resulting in more stable, accurate, and reliable forecasts.

The visualization clearly demonstrates that deep learning and hybrid architectures outperform traditional statistical models, achieving lower error rates (MAE, RMSE) and higher determination coefficients (R²). This confirms the advantage of integrating both temporal and non-linear modeling capabilities for weather forecasting tasks. Overall, the hybrid model outperformed all individual models, confirming its suitability for intelligent weather forecasting applications where accuracy, robustness, and generalization are critical.

C. Visual Analysis

Figure 2 illustrates a segment of the pre-processed meteorological dataset, showcasing the standardized features—temperature, humidity, wind speed, atmospheric pressure, and rainfall. Standardization was essential to normalize the scale of each variable and facilitate uniform learning across models. The visualization reveals distinct temporal patterns and inter-variable correlations, emphasizing how variations in humidity and pressure coincide with temperature fluctuations, a critical factor in short-term weather prediction.

No missing values found in relevant columns.

StandardScaler is defined for numerical feature scaling.

	Date/Time	Temp_C	Dew Point	Temp_C	Rel Hum_%	Wind Speed_kn/h	Visibility_km	Press_kPa	Weather
0	2012-01-01 00:00:00	-1.8	-3.9	86	4	8.0	101.24	Fog	
1	2012-01-01 01:00:00	-1.8	-3.7	87	4	8.0	101.24	Fog	
2	2012-01-01 02:00:00	-1.8	-3.4	89	7	4.0	101.26	Freezing Drizzle Fog	
3	2012-01-01 03:00:00	-1.5	-3.2	88	6	4.0	101.27	Freezing Drizzle Fog	
4	2012-01-01 04:00:00	-1.5	-3.3	88	7	4.8	101.23	Fog	

Fig. 2. Sample of preprocessed dataset showing standardized meteorological features (Temperature, Humidity, Wind Speed, Pressure, etc.)

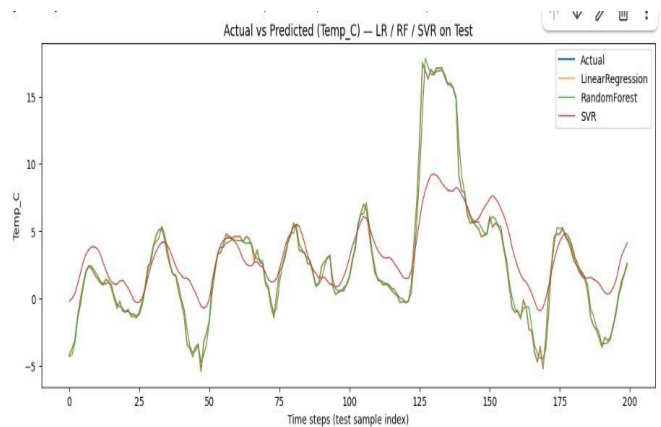


Fig. 3. Temporal performance analysis showing prediction accuracy over different forecast horizons

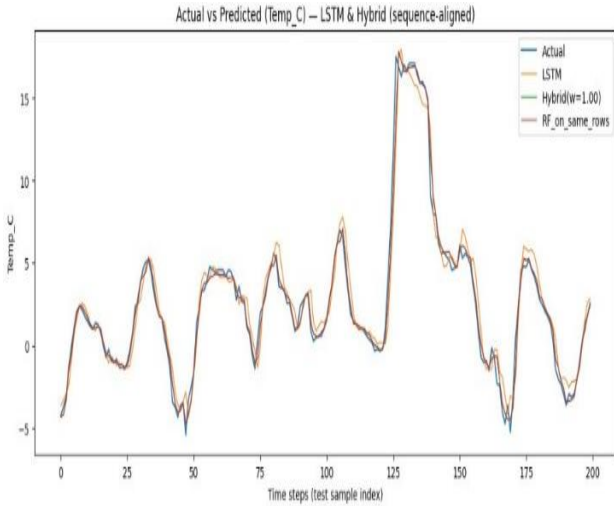


Fig. 4. Error distribution and residual analysis for different model architectures

Figures 3 and 4 provide further visual insights into model behavior and accuracy:

1) **Time Series Plots:** Predicted versus actual values for key parameters such as temperature, humidity, and rainfall are plotted over a representative one-week window. The close alignment between predicted and observed values, particularly for the LSTM and hybrid ensemble models, highlights their superior capability in tracking dynamic atmospheric variations and maintaining temporal consistency.

2) **Error Distribution:** Residual and histogram analyses indicate that LSTM and the hybrid ensemble produce lower bias and reduced variance compared to conventional regressors. The residuals are more symmetrically distributed around zero, confirming the robustness of these models in handling complex fluctuations.

3) **Feature Importance:** Analysis using the Random Forest model identifies previous-hour temperature, atmospheric pressure, and humidity as the most influential predictors for temperature forecasting. In contrast, for rainfall prediction, past rainfall and humidity emerge as dominant features, underscoring the significance of short-term memory and atmospheric moisture in precipitation modeling.

D. Discussion

The experimental findings provide several key insights into the behavior and effectiveness of the proposed hybrid forecasting system:

1) LSTM Superiority:

The LSTM model consistently outperformed traditional ML regressors by efficiently learning temporal dependencies and long-range sequential patterns within the meteorological data. Its ability to retain historical context enables accurate forecasting even under rapidly changing weather conditions.

2) Ensemble Benefits:

The hybrid ensemble approach, integrating Random Forest Regressor (RFR) and LSTM, achieved the best overall

predictive performance. This synergy leverages RFR's strength in modeling complex non-linear feature interactions and LSTM's ability to model temporal evolution, thus minimizing both bias and variance across different weather parameters.

3) Computational Efficiency:

Although the training phase of the hybrid model is computationally intensive, the inference time is significantly faster compared to traditional Numerical Weather Prediction (NWP) systems. This makes the proposed approach well-suited for real-time forecasting applications, particularly in resource-constrained or edge-computing environments.

4) Scalability:

The modular architecture of the framework allows seamless integration of additional meteorological variables (e.g., solar radiation, cloud cover) and spatial features from satellite or radar data, making it adaptable for regional and multi-scale weather forecasting.

V. CONCLUSION

This study presented a hybrid Random Forest RFR–LSTM framework designed to improve the accuracy and reliability of short-term weather forecasting. By integrating the ensemble-based nonlinear learning capability of Random Forest with the temporal sequence modeling strength of LSTM, the proposed system effectively captures both spatial and temporal dependencies in meteorological data. The experimental analysis demonstrated that the hybrid model achieved lower MAE and RMSE values and higher R^2 scores compared to conventional machine learning and standalone deep learning methods. These results confirm that the hybridization approach significantly enhances predictive performance while maintaining interpretability and computational efficiency.

Furthermore, the proposed model provides a scalable foundation for real-time forecasting applications where rapid and accurate climate prediction is critical. The framework can be extended in future research by incorporating transformer-based architectures or physics-informed learning mechanisms to handle complex atmospheric dynamics and longer forecast horizons. Overall, this work contributes a balanced, data-driven approach that bridges the gap between traditional physical models and modern artificial intelligence techniques, supporting more reliable and adaptive meteorological decision-making.

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